AI Medical Scribe

Project Brief

Problem Statement

Doctors spend significant time documenting consultations, preparing referral letters, and recording diagnosis and procedure information. Manual documentation can increase administrative workload and may lead to incomplete or inconsistent records. An AI Medical Scribe can convert doctor–patient conversations into structured draft clinical documentation and suggest relevant medical billing and classification codes for professional review.

Features

- Record or securely import doctor–patient consultation audio with appropriate consent.
- Convert speech into a transcript using a speech-to-text model.
- Generate structured clinical notes, including SOAP notes.
- Extract relevant clinical details such as symptoms, history, examination findings, assessment, and plan.
- Suggest ICD-10-CM diagnosis codes based on documented clinical information.
- Suggest CPT procedure and service codes where applicable.
- Suggest HCPCS Level II codes where applicable.
- Show the rationale and supporting documentation for suggested codes, and flag missing information.
- Generate a draft referral letter containing the reason for referral, relevant history, findings, and requested specialty.
- Allow doctors to edit, approve, and sign notes, code suggestions, and referral letters before finalization.
- Save approved documentation to the authorized patient record or export it in a supported format.
- Maintain access controls and an audit trail for generated and edited documentation.

Suggested Workflow

Consultation audio → speech transcription → structured draft note → ICD/CPT/HCPCS code suggestions → referral letter draft (if needed) → clinician review and approval → save/export.

Suggested Technology

Backend: Python/FastAPI **Speech-to-text:** Whisper or a suitable STT provider **LLM:** a model approved for the deployment **Database:** PostgreSQL **Frontend:** React or another web framework.

Clinical and coding safety: AI-generated notes, referral letters, and ICD/CPT/HCPCS suggestions must be reviewed by an authorized clinician or qualified coding professional before use or submission. Coding requirements vary by jurisdiction and payer; confirm which code sets apply to the Bahrain hospital. Use appropriate consent, access controls, encryption, audit logging, and approved data-handling arrangements. Do not use real patient data in a prototype without authorization.